Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Assignment - 4

Full Marks: 60 *Deadline: 11:59 PM, 25-10-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a zip file which will contain all the .gp, .py, and output files of the task in their respective folders.
3) Each python script file should be placed in its respective task folder (e.g., Task1, Task2..).
4) Your zip file should be named as <Roll No>_<Name>_Assignment_<Lab Assignment No>.zip
5) Example: B24CS055_Shareen_Assignment_14.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1) Evaluate and compute complex mathematical functions accurately using Matplotlib and Gnuplot.
2) Visualize data through smooth 2D heatmaps using Matplotlib and Gnuplot.
3) Customize plots with clear labels, titles, legends, and professional font styles.
4) Create multi-plot layouts using subplots for comparative visualization of functions.
5) Getting familiarized with different plotting tools and using them for visualizing real-life situations.

Task1: [10 Marks]

You are tasked with visualizing Frequency Modulation (FM) by plotting three signals in a single PNG image using Gnuplot via script:

1. Message Signal
2. Carrier Signal
3. Frequency Modulated (FM) Signal

Each plot should be stacked vertically using multiplot mode.

Signal Definitions:

Signal Name	Formula	Parameters
Message Signal	$m(t) = A_m cos(2\Pi f_m t)$	$A_m = 1,\ F_m = 5Hz$
Carrier Signal	$c(t) = A_c cos(2\Pi f_c t)$	$A_c = 2,\ F_c = 50Hz$
Amplitude Modulated Signal	$FM(t) = A_c cos(2\Pi F_c t + k_f sin(2\Pi F_m t)$	$k_f = 5;\ Frequency\ Division\ Index$

Plot Styling Table:

Signal	Color	Line Type	Point Type	Line Width	Title
Message Signal	Blue	1	7	2	“m(t)”
Carrier Signal	Green	2	5	2	“c(t)”
Frequency Modulated Signal	Red	3	9	2	“FM(t)”

Plotting Instructions:

1. Use xrange 0 to 1 to plot signals over 0 to 1 second.
2. Use “set samples 2000” for smooth signal curves.
3. Use set multiplot mode with manual layout by specifying set origin and set size for each plot.
4. Each plot must have a unique line style, point style, and color.
5. Add appropriate titles and axis labels.
6. Save the image as freq_modulation.png with size 1600x900 pixels.

Deliverables:

1. Figure saved as a .png file with the name ‘freq_modulation.png’.
2. .gp file used to create the plot.

Task 2: [20 Marks]

Problem Statement:

Wi-Fi Signal Heatmap Visualization in an Enclosed Room Context:
You are designing a Wi-Fi coverage plan for an 8-meter by 8-meter square room, such as a classroom, office, or lab. The goal is to determine how signal strength from multiple routers behaves spatially in the room, and visualize this in the form of a color-coded heatmap.

Task:

Write a Gnuplot script to simulate and visualize the combined Wi-Fi signal strength across the room. The visualization should reflect how signal strength decreases with distance from each router and accumulates where router signals overlap.

Room Specifications:

Dimensions: 8 meters × 8 meters
Coordinate System: (0,0) is the bottom-left corner of the room, (8,8) is the top-right corner.

Routers and Their Positions:

Router	Position (x,y)	Transmission Power
A	(2,2)	100
B	(6,3)	90
C	(4,6)	80

Each router emits signal strength that decays inversely with the square of the distance from its position, following the inverse-square law.

Signal Model:

For a given point (*x*,*y*), the signal strength from a router at (*cx*,*cy*) with transmission power *P* is modeled by:

$$S(x,y) = \frac{P}{(x-cx)^2+(y-cy)^2+1}$$

The +1 prevents division by zero and simulates attenuation near the source.

The total signal strength at any point is the sum of signal strengths from all three routers:

$$Total(x,y) = S_A(x,y) + S_B(x,y) + S_C(x,y)$$

Output Requirements:

1. Generate a heatmap image (wifi_heatmap_8x8.png) showing signal intensity.
2. The color scale should range from:
    a. Blue (weakest)
    b. Cyan/Green
    c. Yellow
    d. Red (strongest)

3. Include:
    a. Axis labels in meters
    b. A descriptive title
    c. A color bar with signal strength units

Deliverables:

1. Figure saved as a .png file with the name ‘freq_modulation.png’.
2. .gp file used to create the plot.

Task 3: [20 Marks]

Problem Context:

An environmental agency is monitoring air pollution in a 40 × 40 unit industrial zone. Several factories in this region emit pollutants, and the intensity of pollution from each factory decreases exponentially with distance. Your task is to model this pollution distribution and visualize it as a heatmap using Python's matplotlib.

Pollution Model:

The pollution intensity: $P(x,y)$ at any location (x,y) is given by:

$$P(x,y) = \sum_{i=1}^{N} I_a \cdot e^{k \cdot d_i(x,y)}$$

Where:

- I_i is the emission intensity of factory
- $d_i(x,y)$ is the Euclidean distance from point (x,y) to factory i,
- $k = 0.1$ is the decay rate constant.
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Factory Emission Data:

Use the following list of factories, where each factory is defined by its (x position, y position, emission intensity):

factories = [

(8, 8, 100),

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    (30, 10, 120),
    (20, 35, 90),
    (10, 25, 110),
    (35, 30, 80)
]
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Your Tasks:

1. Set up a 2D grid of coordinates covering the region from 0 to 40 units on both axes. Use 200 points in each direction.
2. Define a function distance(x1, y1, x2, y2) to compute Euclidean distance.
3. Define a function pollution(x, y) that computes the total pollution level at each point in the grid by summing contributions from all factories.
4. Use NumPy to calculate pollution intensity across the entire grid efficiently.
5. Create a heatmap using matplotlib to visualize the pollution intensity:
6. Use a custom colormap that transitions from blue → cyan → green → yellow → red to indicate increasing pollution levels.
7. Set the figure size to 30 cm × 14 cm.
8. Use the Times New Roman font for all labels and titles.
9. Display a colorbar with the label "Pollution Intensity (arb. units)".
10. Mark the positions of the factories with red circles on the heatmap.
11. Set axis labels and a descriptive title.

Expected Output:

A smooth, colorful heatmap showing areas of high and low pollution intensity across the zone. Factory locations should be clearly marked. Generate heatmap image factory_pollution.pdf and save it.

Deliverables:

1. Figure saved as a .pdf file with the name ‘factory_pollution.pdf.
2. .py file used to create the plot.

Task 4: [10 Marks]

Problem Context:
A household tracks its monthly spending across different categories to understand where most of its income goes. Your task is to create a pie chart to represent this expenditure distribution using Python's matplotlib.

Data Description:
The household's expenses are broken down as follows:
categories = ['Rent', 'Groceries', 'Utilities', 'Transportation', 'Entertainment', 'Savings']
expenses = [1200, 450, 200, 150, 300, 700]

Your Tasks:

1. Define the categories and expenses as two Python lists.
2. Choose a distinct color for each slice of the pie chart.
3. Highlight the 'Rent' slice by exploding it slightly from the chart.
4. Add percentage values on each slice (1 decimal place).
5. Add a legend beside the pie chart.
6. Add a meaningful title to the plot:
    a. "Monthly Household Expenditure Breakdown"
    b. Rotate the start angle of the chart by 90 degrees for better readability.
7. Set the figure size to 8×8 inches.
8. Add a shadow for visual depth.
9. Save the final chart as a PDF named monthly_expenses.pdf.

Expected Output:
A clear and visually appealing pie chart showing the proportion of each category of household spending. The 'Rent' slice should be slightly separated from the rest, and a legend should be shown.

Deliverables:

1. Figure saved as a .pdf file with the name ‘monthly_expenses.pdf’.
2. .py file used to create the plot.